Computer Organization (CSD206): Tutorial 10

For this assignment, you will write questions based on Computer Memory.

Questions

1. Discuss the importance of having a memory hierarchy in a computer system.
2. Discuss the strategies for exploiting spatial locality and temporal locality.
3. Consider the following code:

```
for (m = 15; m > 0; m--)
    while (n < 27)
        a[m] = a[m]+ n
```

   a) Give one example of the spatial locality in the code.
   b) Give one example of the temporal locality in the code.

4. Suppose that the processor has access to two levels of memory. Level 1 contains 1000 words and has an access time of 0.01 μs; level 2 contains 100,000 words and has an access time of 0.1 μs. Assume that if a word to be accessed is in level 1, then the processor accesses it directly. If it is in level 2, then the word is first transferred to level 1 and then accessed by the processor. For simplicity, we ignore the time required for the processor to determine whether the word is in level 1 or level 2. Suppose 95% of the memory accesses are found in level 1. Calculate the average time to access a word.

5. A computer has a cache, main memory, and a disk used for virtual memory. If a referenced word is in the cache, 20 ns are required to access it. If it is in main memory but not in the cache, 60 ns are needed to load it into the cache, and then the reference is started again. If the word is not in main memory, 12 ms are required to fetch the word from disk, followed by 60 ns to copy it to the cache, and then the reference is started again. The cache hit ratio is 0.9 and the main memory hit ratio is 0.6. What is the time in nanoseconds required to access a referenced word on this system?

6. Assume a processor having a memory cycle time of 300 ns and an instruction processing rate of 1 MIPS. On average, each instruction requires one bus memory cycle for instruction fetch and one for the operand it involves. Calculate the utilization of the bus by the processor. Suppose the processor is equipped with an instruction cache and the associated hit ratio is 0.5. Determine the impact on bus utilization.

7. Consider a single-level cache with an access time of 2.5 ns, a line size of 64 bytes, and a hit ratio of $H = 0.95$. Main memory uses a block transfer capability that has a first-word (4 bytes) access time of 50 ns and an access time of 5 ns for each word thereafter.
   a) What is the access time when there is a cache miss? Assume that the cache waits until the line has been fetched from main memory and then re-executes for a hit.
   b) Suppose that increasing the line size to 128 bytes increases the H to 0.97. Does this reduce the average memory access time?